Finite-Support Stabilization for *Lights Out* Nullity

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Abstract

Let $d(n)$ be the nullity over $\mathbb{F}_2$ of the $n \times n$ bounded *Lights Out* grid. For every fixed finite set S of odd primes, we give explicit thresholds $C_p(S)$ such that

$$d\left(\prod_{p\in S} p^{e_p} - 1\right) = d\left(\prod_{p\in S} p^{\min(e_p, C_p(S))} - 1\right).$$

Thus the nullity is eventually constant in each prime exponent. In particular, for every odd integer a, the sequence $d(a^k - 1)$ is eventually constant. We also construct, for every $K \geq 3$, an odd integer whose true smallest stabilization exponent is exactly K. The thresholds explain the observed transitions for $a = 21$ and $a = 57$, and reduce previously observed mixed-prime formulas to small exact polynomial computations.

1 Polynomial and order preliminaries

Let $F_n(x) \in \mathbb{F}_2[x]$ be the Fibonacci polynomials

$$F_0(x) = 0, \qquad F_1(x) = 1, \qquad F_n(x) = xF_{n-1}(x) + F_{n-2}(x).$$

Sutner's formula gives

$$d(n) = \deg \gcd(F_{n+1}(x), F_{n+1}(x+1)) \tag{1}$$

[Sut00].

For an odd integer $M > 1$, define the exact-order stratum

$$R_M = \left\{\zeta + \zeta^{-1} : \operatorname{ord}(\zeta) = M\right\}.$$

If N is odd, the distinct roots of F_N are the disjoint union

$$\bigcup_{\substack{M\mid N\\ M>1}} R_M. \tag{2}$$

Moreover, F_N is the square of a square-free polynomial, so every root has multiplicity two [HMP04].

For odd M, let

$$\rho_2(M) = \min\{r > 0 : 2^r \equiv \pm 1 \pmod{M}\}.$$

If $\alpha \in R_M$, then

$$[\mathbb{F}_2(\alpha) : \mathbb{F}_2] = \rho_2(M). \tag{3}$$

Fix a finite set S of odd primes. For each $p \in S$, set

$$h_p = \operatorname{ord}_p(2), \qquad c_p = v_p(2^{h_p} - 1), \qquad a_p(S) = \max_{q\in S} v_p(h_q), \qquad C_p(S) = c_p + a_p(S). \tag{4}$$

The term c_p is the prime-power plateau length from the one-prime argument. The additional term $a_p(S)$ measures how much p-power divisibility can enter through the orders belonging to the other primes in S.

For an odd integer N supported on S, define its S-truncation by

$$N^\flat = \prod_{p \in S} p^{\min(v_p(N), C_p(S))}. \tag{5}$$

2 Order growth after truncation

Lemma 2.1. *Let Q be an odd integer supported on S, let $Q^\flat$ be its truncation, and set*

$$m = \frac{Q}{Q^\flat}.$$

Then

$$\mathrm{ord}_Q(2) = m \, \mathrm{ord}_{Q^\flat}(2)\,.$$

Proof. Write $u_p = v_p(Q)$. The Chinese remainder theorem and the prime-power order formula give

$$\mathrm{ord}_Q(2) = \operatorname*{lcm}_{\substack{p \in S \\ u_p > 0}} \Big(h_p \, p^{\max(0, u_p - c_p)} \Big)\,. \tag{6}$$

First consider a prime $\ell \notin S$. Its valuation in (6) comes only from the fixed integers h_p. Truncation changes no prime support, so

$$v_\ell(\mathrm{ord}_Q(2)) = v_\ell\big(\mathrm{ord}_{Q^\flat}(2)\big)\,.$$

Now fix $p \in S$. The p-adic valuation in (6) is the maximum of

$$\max(0, u_p - c_p) \qquad \text{and} \qquad \{v_p(h_q) : q \in S,\ q \neq p,\ u_q > 0\}.$$

Every term in the second set is at most $a_p(S)$. If $u_p \leq C_p(S)$, truncation leaves u_p unchanged, and truncating the other exponents does not change their fixed factors h_q. The p-adic valuation of the order is therefore unchanged.

If $u_p > C_p(S) = c_p + a_p(S)$, then

$$u_p - c_p > a_p(S),$$

so the contribution from the p-power component strictly dominates all cross-prime contributions. Hence

$$v_p(\mathrm{ord}_Q(2)) = u_p - c_p.$$

At the truncated exponent $C_p(S)$, the corresponding valuation is $a_p(S)$, and therefore

$$v_p(\mathrm{ord}_Q(2)) - v_p\big(\mathrm{ord}_{Q^\flat}(2)\big) = u_p - C_p(S) = v_p(m).$$

Comparing valuations at every prime proves the result. ∎

3 Finite-support stabilization

Theorem 3.1 (Finite-support truncation)**.** *Let S be a finite set of odd primes and let N be an odd positive integer supported on S. Then*

$$d(N-1) = d(N^\flat - 1),$$

where $N^\flat$ is defined by (5)*.*

Proof. Suppose α and $\alpha+1$ are roots of F_N. By (2), there are divisors $M, L > 1$ of N such that

$$\alpha \in R_M, \qquad \alpha+1 \in R_L.$$

Because $\mathbb{F}_2(\alpha) = \mathbb{F}_2(\alpha+1)$, equation (3) gives

$$\rho_2(M) = \rho_2(L). \tag{7}$$

Write $j_p = v_p(M)$ and $\ell_p = v_p(L)$. Fix $p \in S$ and suppose $j_p > C_p(S)$. As in the proof of Lemma 2.1, the p-part belonging to p^{j_p} dominates every cross-prime contribution, so

$$v_p(\mathrm{ord}_M(2)) = j_p - c_p > a_p(S).$$

Signed order differs from ordinary order by at most a factor of two, so their valuations at the odd prime p agree. Equation (7) therefore forces

$$v_p(\mathrm{ord}_L(2)) = j_p - c_p.$$

This valuation is larger than $a_p(S)$, so it cannot arise from any cross-prime order h_q. Consequently $\ell_p > C_p(S)$ and

$$\ell_p - c_p = j_p - c_p, \qquad \text{hence} \qquad \ell_p = j_p.$$

By symmetry, every exponent above its threshold occurs with the same value in M and L.

Set

$$Q = \mathrm{lcm}(M, L), \qquad Q^\flat = \prod_{p \in S} p^{\min(v_p(Q), C_p(S))}, \qquad m = \frac{Q}{Q^\flat}.$$

If $m = 1$, then $M, L \mid N^\flat$ and there is nothing to prove. Suppose $m > 1$.

Choose a primitive Qth root θ. There are exponents a, b such that

$$\alpha = \theta^a + \theta^{-a}, \qquad \alpha + 1 = \theta^b + \theta^{-b},$$

where θ^a and θ^b have exact orders M and L. Thus θ is a root of

$$P(x) = 1 + x^a + x^{Q-a} + x^b + x^{Q-b}. \tag{8}$$

Every prime dividing m occurs above its threshold in Q. The preceding rigidity argument says that its full exponent in Q occurs in both M and L. Since θ^a has exact order M, we have

$$\gcd(Q, a) = \frac{Q}{M};$$

no prime dividing m divides Q/M, and no such prime can divide the remaining unit factor in a. Hence $\gcd(a, m) = 1$. Likewise $\gcd(b, m) = 1$. Because $m \mid Q$, only the constant exponent in (8) is divisible by m.

The element θ^m has exact order $Q^\flat$. Let $f(x)$ be its minimal polynomial over $\mathbb{F}_2$. Lemma 2.1 gives

$$\deg f(x^m) = m\, \mathrm{ord}_{Q^\flat}(2) = \mathrm{ord}_Q(2)\,,$$

the degree of the minimal polynomial of θ. Since θ is a root of $f(x^m)$, the polynomial $f(x^m)$ is its minimal polynomial. Therefore $f(x^m) \mid P(x)$.

Decompose (8) uniquely by exponents modulo m:

$$P(x) = \sum_{r=0}^{m-1} x^r P_r(x^m).$$

Divisibility by $f(x^m)$ implies that $f(y)$ divides every $P_r(y)$. But the preceding coprimality shows that $P_0(y) = 1$, a contradiction. Thus $m = 1$, and every common root of $F_N(x)$ and $F_N(x+1)$ already occurs for $F_{N^\flat}(x)$ and $F_{N^\flat}(x+1)$.

The converse inclusion follows from $N^\flat \mid N$ and the Fibonacci-polynomial divisibility property. Because both indices are odd, every root has multiplicity two. Equation (1) now gives

$$d(N-1) = d(N^\flat - 1).$$

■

Corollary 3.2 (Eventual constancy)**.** *For every odd positive integer A, there exists k_0 such that*

$$d(A^k - 1) = d(A^{k_0} - 1) \qquad \text{for every } k \geq k_0.$$

More explicitly, write

$$A = \prod_{p \in S} p^{b_p}$$

and set

$$k_0 = \max_{p \in S} \left\lceil \frac{C_p(S)}{b_p} \right\rceil.$$

Then this k_0 has the required property.

Proof. For every $k \geq k_0$, the truncation of A^k is the fixed integer

$$B_S = \prod_{p \in S} p^{C_p(S)}.$$

Theorem 3.1 gives

$$d(A^k - 1) = d(B_S - 1).$$

The same theorem applied to A^{k_0} gives

$$d(A^{k_0} - 1) = d(B_S - 1),$$

which proves the claim. ■

4 Arbitrarily late stabilization

The sufficient cutoff in Corollary 3.2 can be arbitrarily large. More strongly, the actual smallest stabilization exponent is unbounded.

Theorem 4.1. *For every integer $K \geq 3$, there exists an odd positive integer A_K such that K is the smallest integer satisfying*

$$d(A_K^k - 1) = d(A_K^K - 1) \qquad \text{for every } k \geq K.$$

Proof. Set

$$r = 3^{K-1}, \qquad B = 2^r + 1.$$

The lifting-the-exponent lemma gives

$$v_3(B) = v_3(2+1) + v_3(r) = K.$$

Write

$$B = 3^K Q, \qquad \gcd(Q, 3) = 1,$$

and define

$$A_K = 3Q = \frac{2^{3^{K-1}} + 1}{3^{K-1}}. \tag{9}$$

Let S be the prime support of A_K. If $q \mid Q$, then $2^r \equiv -1 \pmod{q}$, so

$$h_q = \mathrm{ord}_q(2) \mid 2r \qquad \text{but} \qquad h_q \nmid r.$$

Since r is a power of 3, this means

$$h_q = 2 \cdot 3^j \qquad \text{for some } 0 \le j \le K-1.$$

Zsigmondy's theorem provides a primitive prime divisor of $2^{2r} - 1$ because $2r \ge 18$ [Zsi92]. Such a prime has order exactly $2r$, so it divides $2^r + 1$. It is not 3, whose order is 2, and therefore divides Q and belongs to S. Hence the maximum value of $v_3(h_q)$ over $q \in S$ is $K - 1$. Therefore

$$C_3(S) = c_3 + a_3(S) = 1 + (K-1) = K.$$

Now let $q > 3$ divide Q. Every h_t for $t \in S$ divides $2r = 2 \cdot 3^{K-1}$, so

$$a_q(S) = 0.$$

Furthermore, $h_q \mid 2r$ and $q \nmid 2r/h_q$. The lifting-the-exponent lemma gives

$$c_q = v_q(2^{h_q} - 1) = v_q(2^{2r} - 1).$$

Since $q \mid 2^r + 1$ and $q \nmid 2^r - 1$,

$$c_q = v_q(2^r + 1) = v_q(B).$$

Consequently

$$C_q(S) = v_q(B) \qquad (q \mid Q).$$

The exponent of 3 in A_K is one, while the exponent of every $q \mid Q$ is already $C_q(S)$. Hence Theorem 3.1 gives

$$d(A_K^k - 1) = d(3^{\min(k,K)} Q - 1). \tag{10}$$

In particular the sequence is constant for $k \ge K$.

It remains to show that it has not stabilized earlier. For $k < K$, set $N_k = 3^k Q$. Then

$$N_k = \frac{B}{3^{K-k}} \le \frac{B}{3},$$

and the elementary bound $d(n) \le n$ gives

$$d(A_K^k - 1) = d(N_k - 1) \le N_k - 1 < \frac{B}{3}. \tag{11}$$

On the other hand, the reduction and Kloosterman evaluations proved in `tex/upper_bound/upper_bound.tex` give

$$d(2^r) = G(2^r + 1) - 2$$

because $3 \mid 2^r + 1$, and

$$G(2^r + 1) = \frac{2^r + 1 + \mathcal{K}_r}{2}.$$

Consequently

$$d(B - 1) = d(2^r) = \frac{2^r - 3 + \mathcal{K}_r}{2},$$

where $\mathcal{K}_r$ is the binary Kloosterman sum. Weil's bound $|\mathcal{K}_r| \le 2\sqrt{2^r}$ [Wei48] yields

$$d(B-1) \ge \frac{2^r - 3 - 2\sqrt{2^r}}{2}.$$

Because $r \ge 9$,

$$2^r - 6\sqrt{2^r} - 11 > 0,$$

which is equivalent to

$$\frac{2^r - 3 - 2\sqrt{2^r}}{2} > \frac{2^r + 1}{3} = \frac{B}{3}.$$

Combining this with (11) shows

$$d(A_K^k - 1) < d(A_K^K - 1) \qquad (k < K).$$

Thus the true smallest stabilization exponent is exactly K. ■

Example 4.2. *For $K = 3$, construction (9) gives*

$$A_3 = \frac{2^9 + 1}{3^2} = 57,$$

recovering the sequence

$$0,\ 36,\ 252,\ 252, \ldots.$$

For $K = 4$ it gives

$$A_4 = \frac{2^{27} + 1}{3^3} = 4\,971\,027 = 3 \cdot 19 \cdot 87\,211,$$

whose true stabilization exponent is 4.

5 Examples

5.1 The family $3^n 7^m$

We have

$$h_3 = 2, \quad c_3 = 1, \qquad h_7 = 3, \quad c_7 = 1.$$

Since $v_3(h_7) = 1$, the thresholds for $S = \{3, 7\}$ are

$$C_3(S) = 2, \qquad C_7(S) = 1.$$

Therefore

$$d(3^n 7^m - 1) = d\Big(3^{\min(n,2)} 7^{\min(m,1)} - 1\Big). \tag{12}$$

The finite calculations

$$d(3 \cdot 7 - 1) = d(20) = 0, \qquad d(3^2 \cdot 7 - 1) = d(62) = 24$$

now prove, for positive m,

$$d(3^n 7^m - 1) = \begin{cases} 0 & n = 1, \\ 24 & n \ge 2. \end{cases}$$

When $m = 0$, prime-power stabilization gives $d(3^n - 1) = 0$. Thus the second range in the corresponding piecewise formula is $n \ge 2$, not $n \ge 1$.

For $A = 21$, Corollary 3.2 gives $k_0 = 2$, and hence

$$d(21^k - 1) = 24 \qquad (k \ge 2).$$

5.2 The family 3^n19^m

Here

$$h_{19} = 18, \qquad v_3(h_{19}) = 2,$$

so

$$C_3(\{3, 19\}) = 3, \qquad C_{19}(\{3, 19\}) = 1.$$

Consequently

$$d(3^n19^m - 1) = d\Big(3^{\min(n,3)}19^{\min(m,1)} - 1\Big).$$

The exact finite values

$$d(57 - 1) = 0, \qquad d(171 - 1) = 36, \qquad d(513 - 1) = 252$$

prove, for positive m,

$$d(3^n19^m - 1) = \begin{cases} 0 & n = 1, \\ 36 & n = 2, \\ 252 & n \geq 3. \end{cases}$$

For $A = 57$, the sufficient stabilization exponent is $k_0 = 3$:

$$d(57^k - 1) = 252 \qquad (k \geq 3).$$

5.3 Further two-prime families

The same thresholds reduce the remaining positive-exponent two-prime formulas suggested by computation to the following finite GCD calculations:

$$\begin{array}{lll}
d(15-1) = 4, & d(33-1) = 20, & d(39-1) = d(117-1) = 0, \\
d(51-1) = 8, & d(35-1) = 4, & d(55-1) = d(275-1) = 4, \\
d(65-1) = 28, & d(85-1) = 12, & d(95-1) = 4, \\
d(77-1) = 0, & d(91-1) = 0, & d(119-1) = 8, \\
d(133-1) = 0, & d(143-1) = 0, & d(187-1) = 8, \\
d(209-1) = 0, & d(221-1) = 8, & d(247-1) = 0, \\
d(323-1) = 8. & &
\end{array}$$

These exact checks are performed by `code/multiprime_stabilization_check.py`. Together with Theorem 3.1, they prove all of those formulas for positive prime exponents.

Remark 5.1. *The thresholds $C_p(S)$ are sufficient but need not be minimal. For example, $C_5(\{5, 11\}) = 2$ because $5 \mid \mathrm{ord}_{11}(2)$, but the two finite values at exponents one and two are both 4. Thus the nullity in that family is already constant from the first exponent.*

References

[HMP04] Markus Hunziker, António Machiavelo, and Jihun Park, *Chebyshev polynomials over finite fields and reversibility of σ-automata on square grids*, Theoretical Computer Science **320** (2004), no. 2, 465–483.

[Sut00] Klaus Sutner, *σ-automata and chebyshev-polynomials*, Theoretical Computer Science **230** (2000), no. 1, 49–73.

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